D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

Visualization Tool for Electric Vehicle Charge and Range Analysis

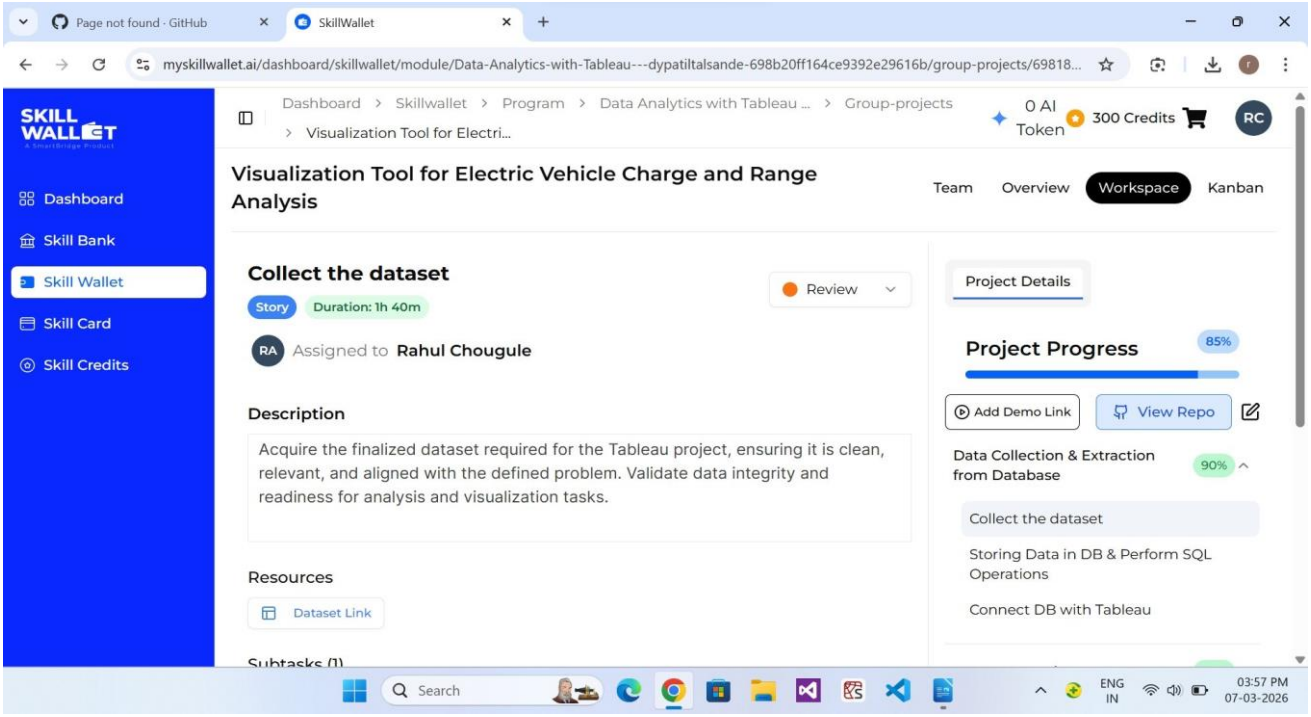
TEAM MEMBERS

1) Pranesh Kamble
2) Vrushabh Kole
3) Pradnya Gatage
4) Habib Mullani

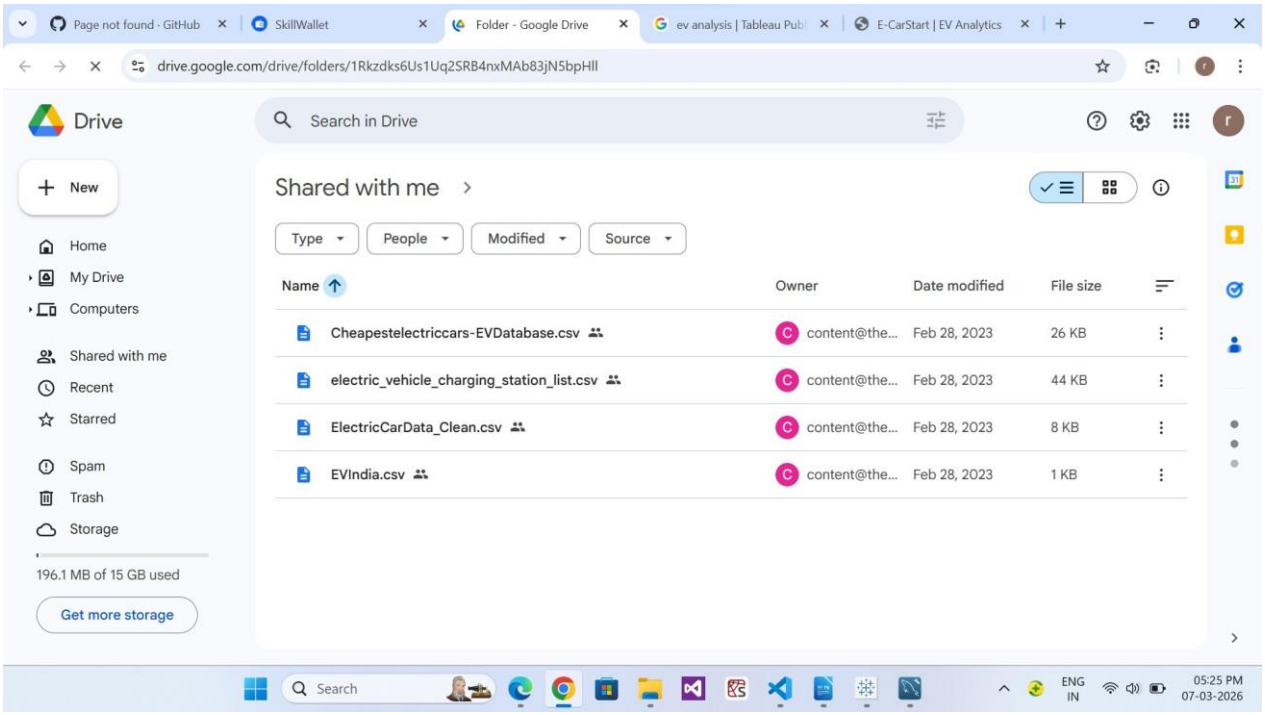
1) Data Collection & Extraction from Database:

- **Collect the dataset:**

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link** .



2) From the given dataset link drive opens , then download the dataset



❖ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File" On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)

Text File (for .csv)
Click the correct
option.

Step 2: Select Your File

Browse your computer.
Select your dataset file.
Click
Open.

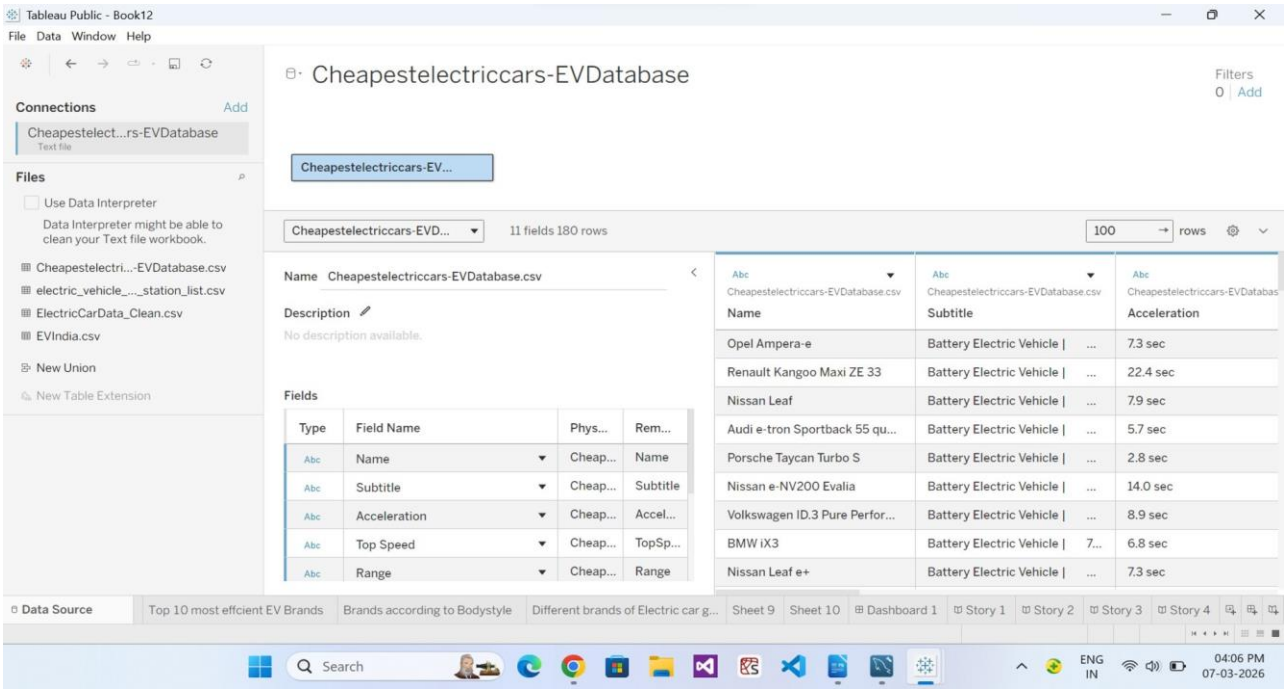
Step 3: Data Source Page

Your sheet names will appear on the left side.
Drag the sheet to the centre area.
Tableau will show a preview of your data.

2) Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.

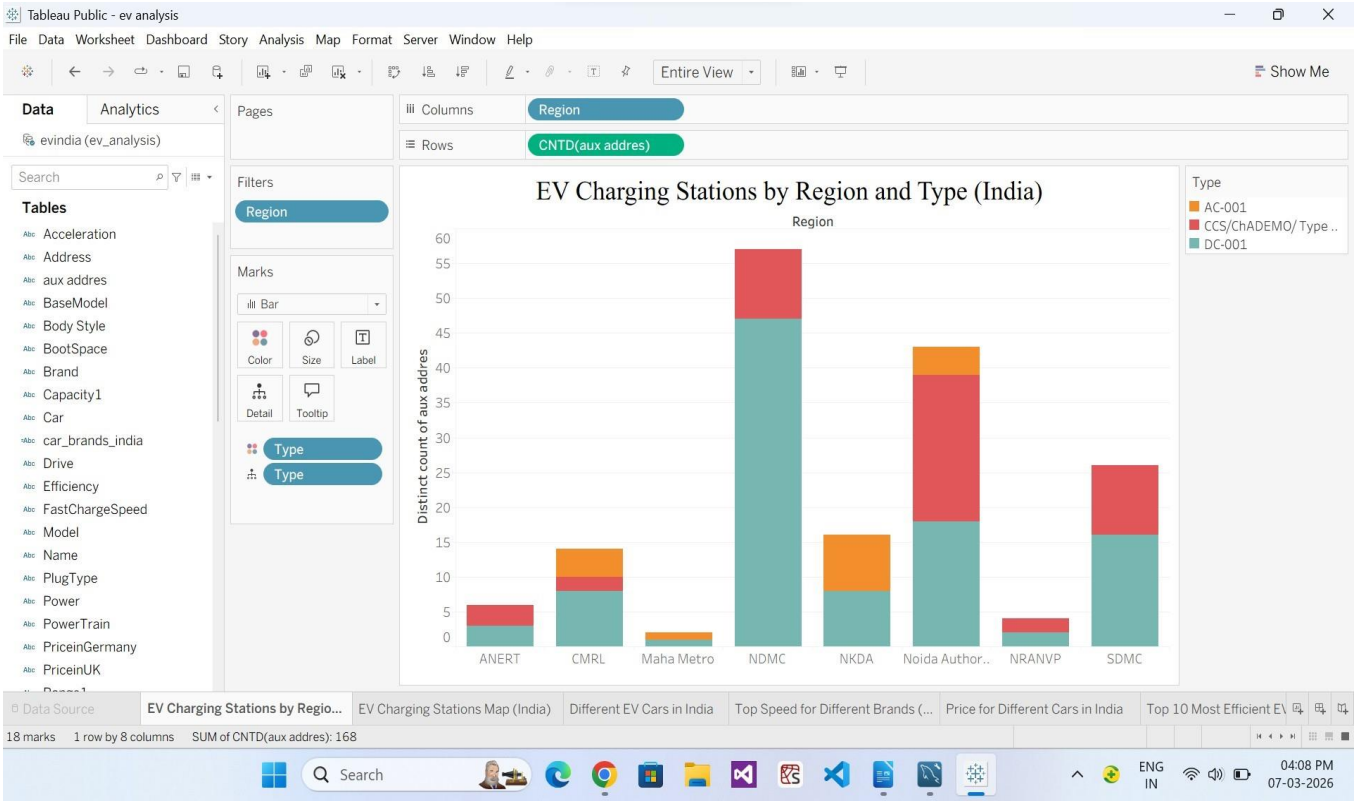


3) Data Visualization:

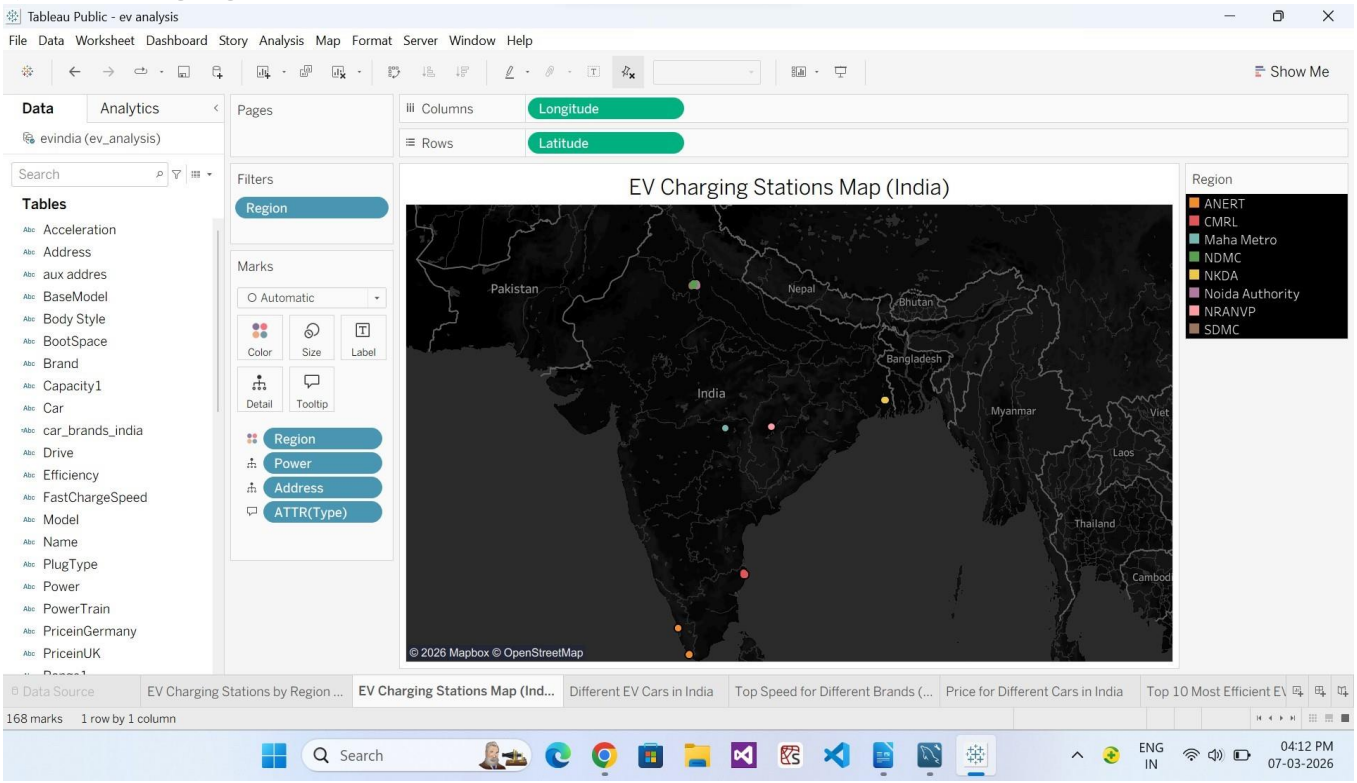
Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖No of Unique Visualizations:

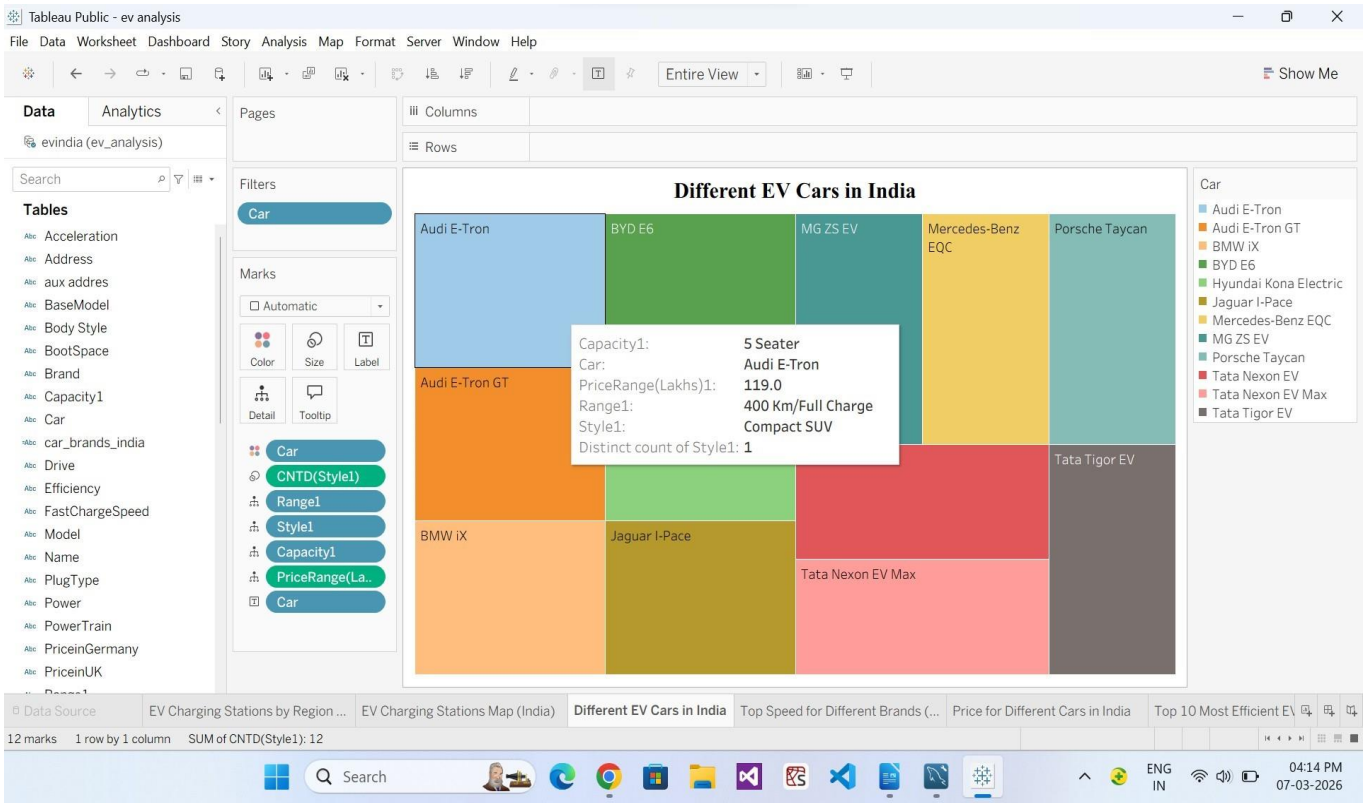
1) charging stations in India, categorized by region and type.



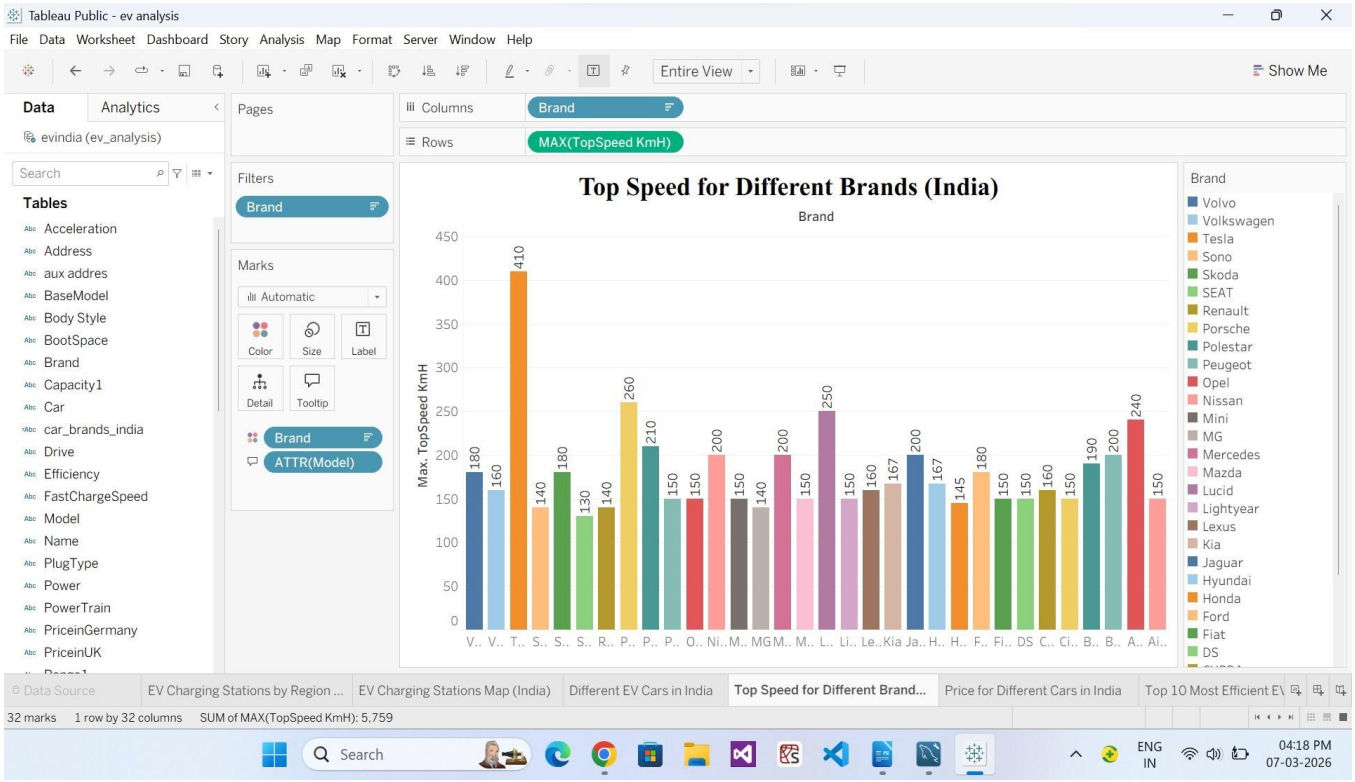
2) EV charging stations across India,



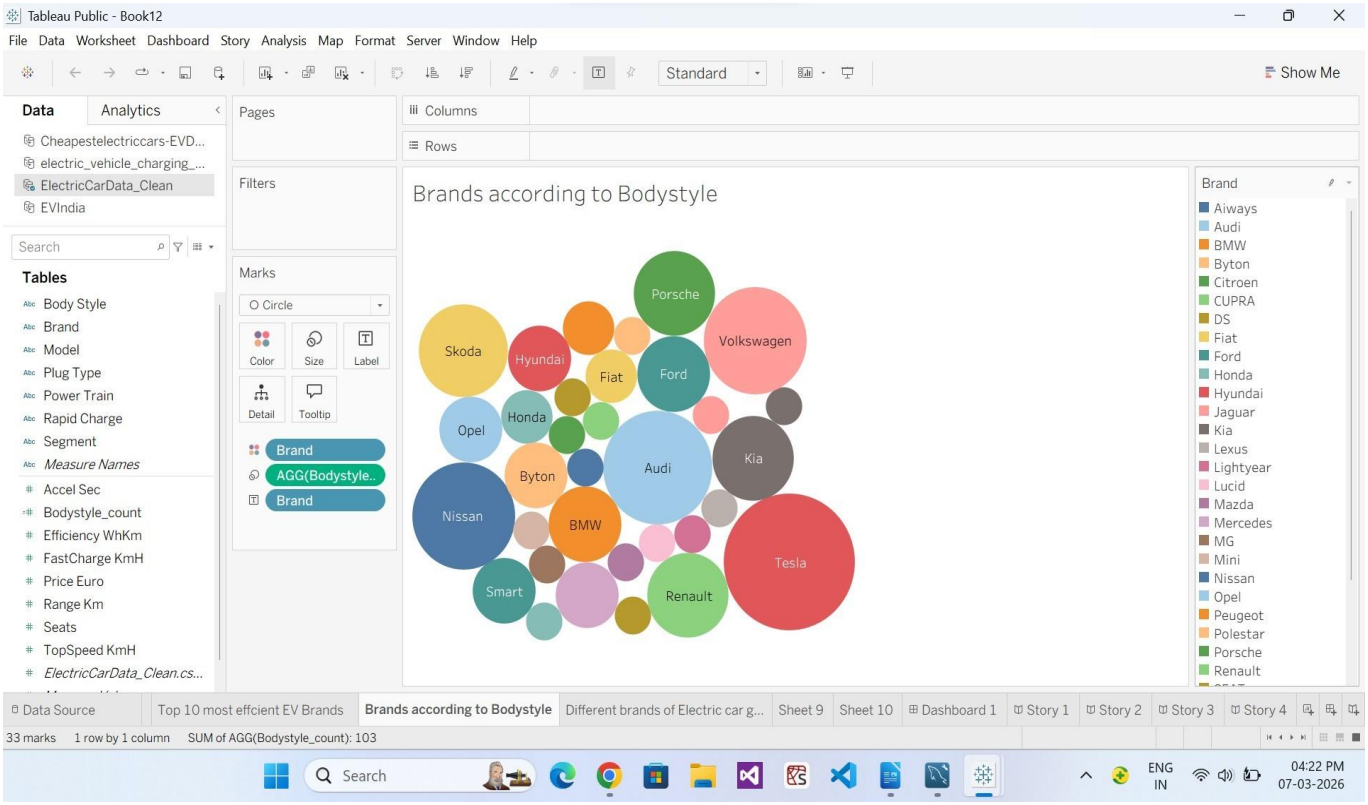
3) different EV car models available in India



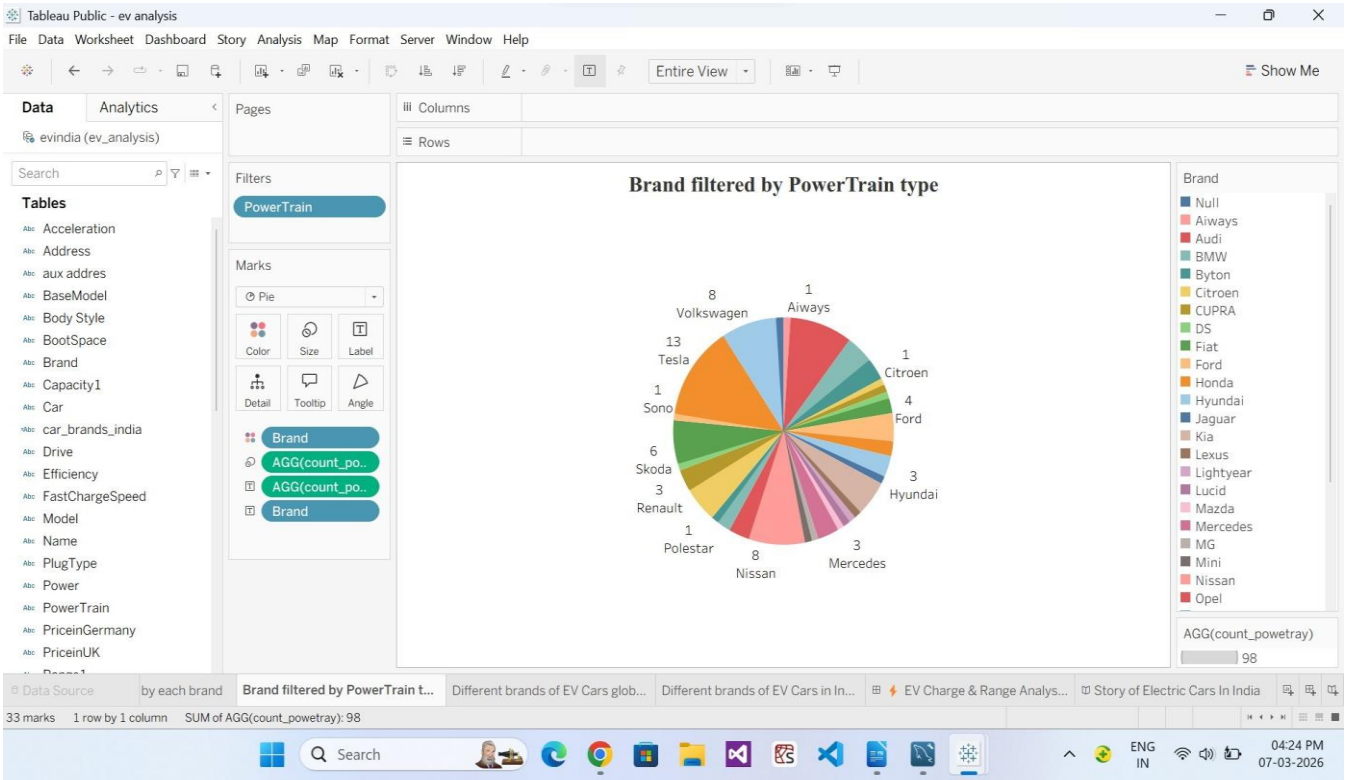
4) Top Speed for Different Brands



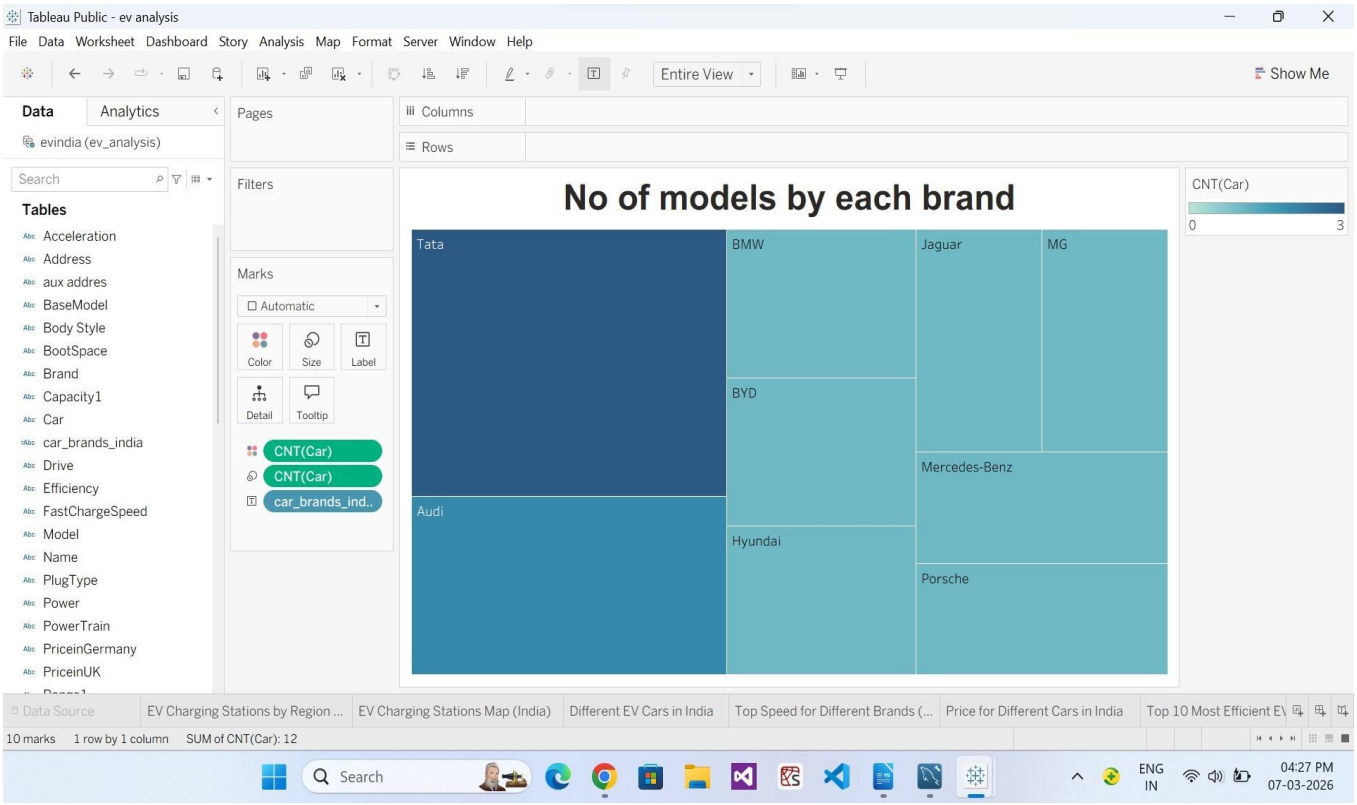
5) brands according to Boodystyle



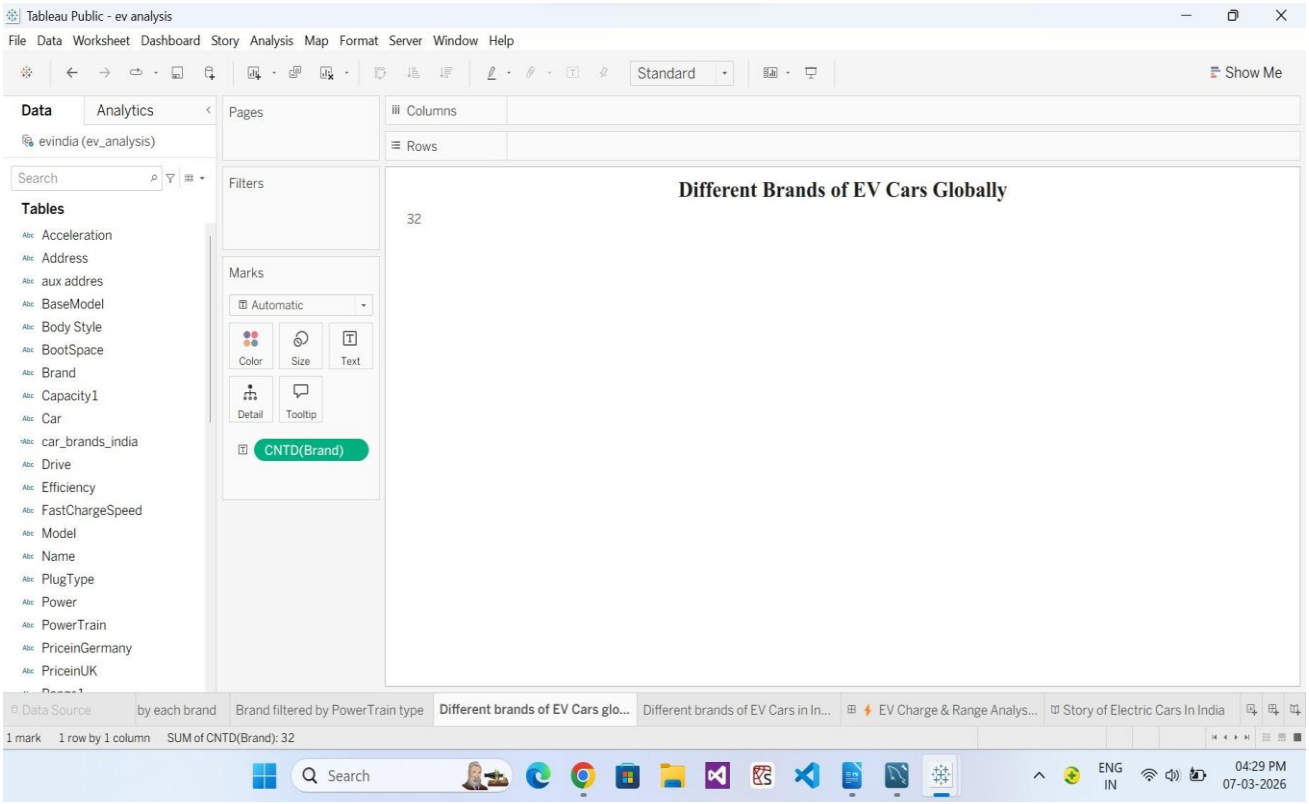
6) Brand filtered by PowerTrain type



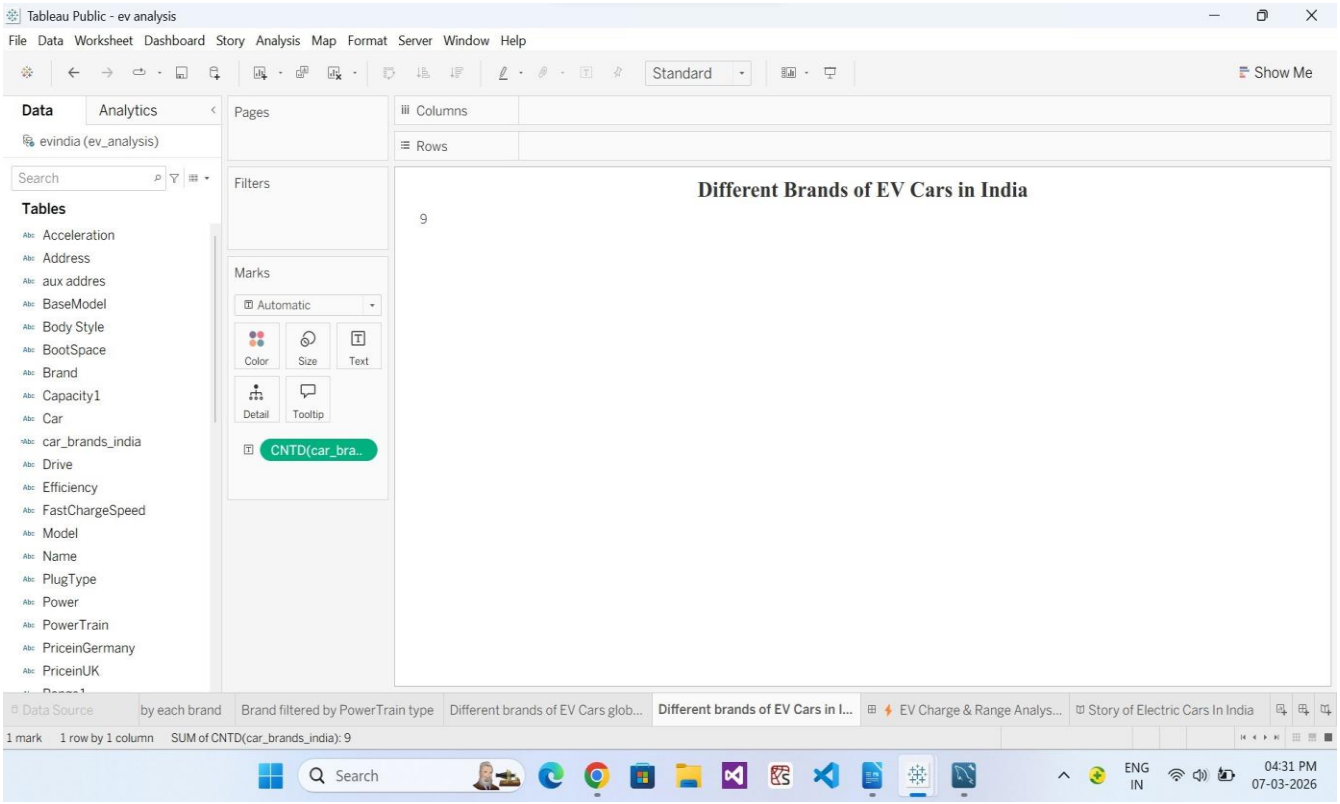
7) No of models by each brand



8) Summary card for Different brands of EV Cars globally



9) Summary card for Different brands of EV Cars India



4) Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

❖Creating Dashboard 1 — Visualization Tool for Electric Vehicle Charge and Range Analysis

❖Step 1: Create a New Dashboard

a) At the bottom of Tableau, click the New Dashboard button (the icon next to the sheet tabs).
b) A blank dashboard canvas will open.
c) The dashboard will be named Dashboard 1 by default.

Step 2: Set the Dashboard Size

a) On the left panel under Size, click the dropdown.
b) Select Fixed Size → choose Desktop Browser (1000 x 800) or select Automatic for responsive layout.

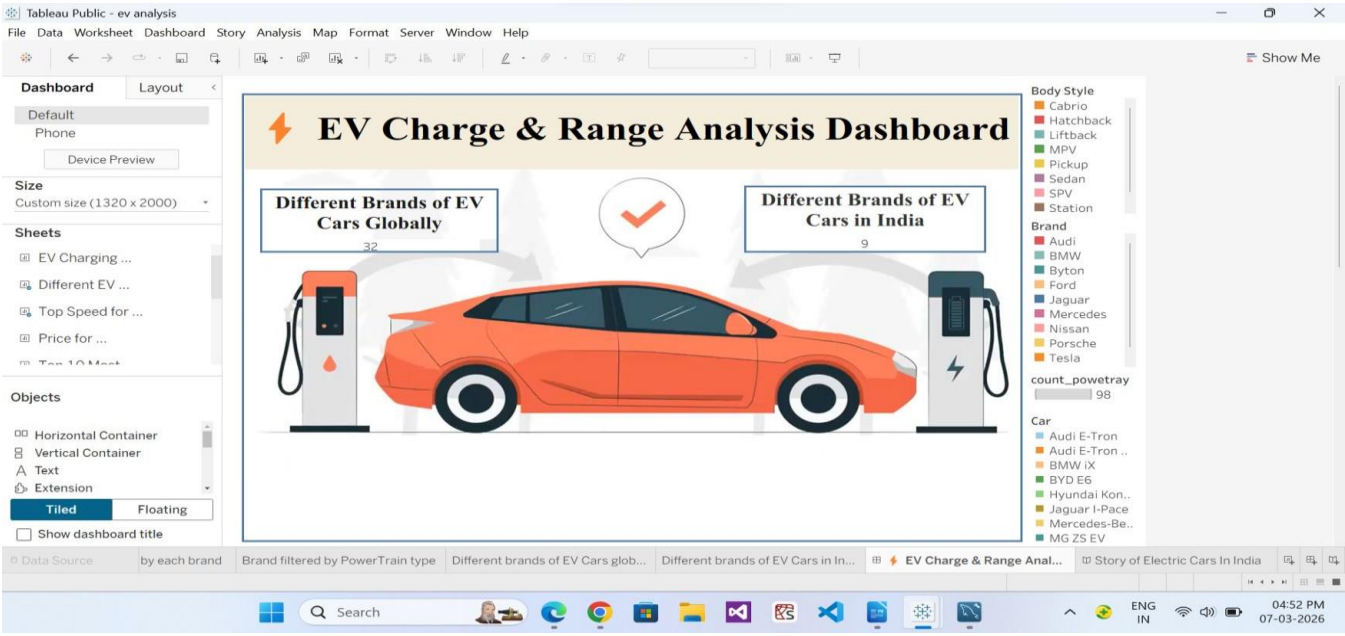
Step 3: Add Sheet 1

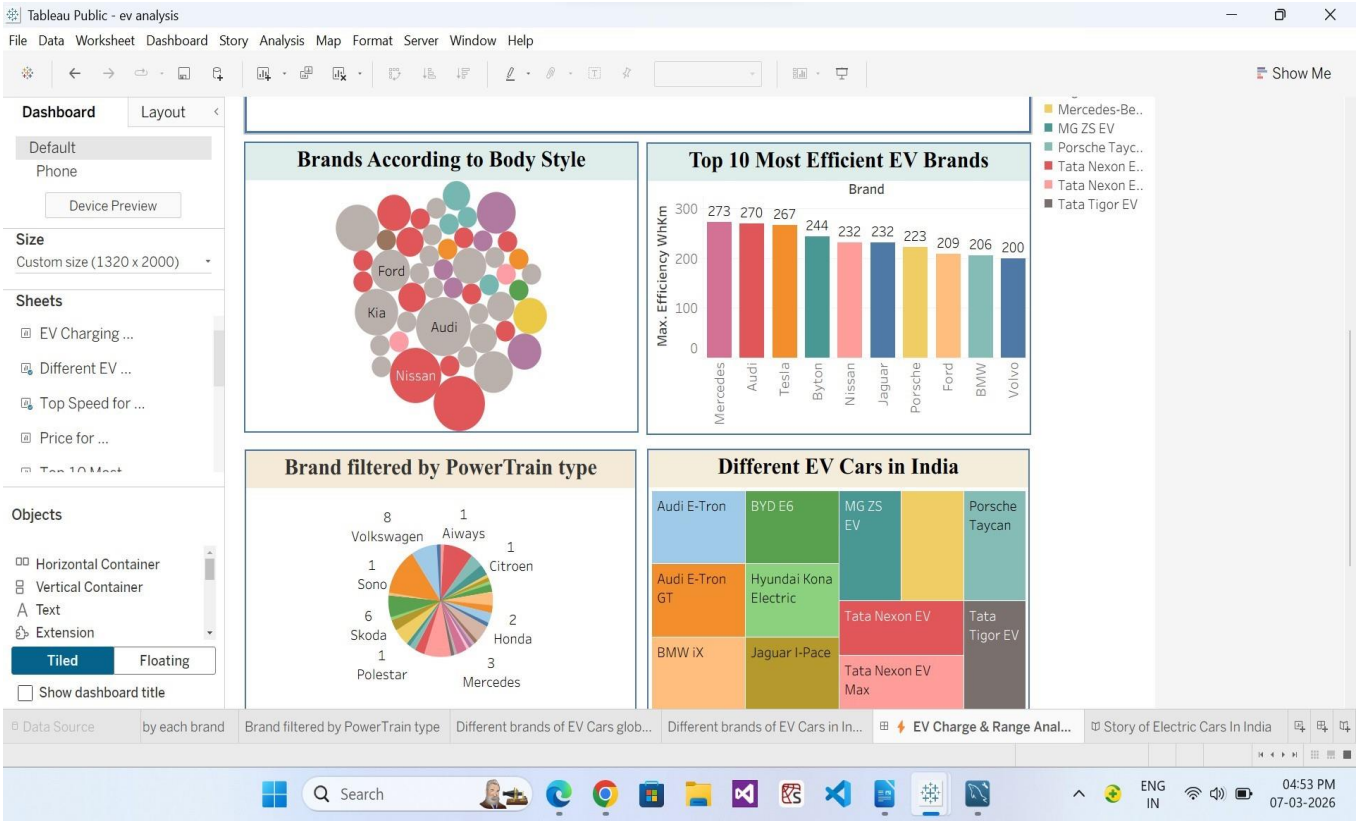
a) From the left Sheets panel, drag Sheet 1 into the centre canvas.
b) Drop it in the top-left area of the dashboard.
c) On the right side of the sheet a downward arrow appears — click it and select Floating option if you want free positioning, or leave as Tiled for auto layout.
d) Resize the sheet by dragging its edges to fit the top-left quadrant.

Step 4: Add Sheet 2 —
a) From the left Sheets panel, drag Sheet 2 into the canvas.
b) Drop it in the top-right area of the dashboard.
c) Resize to fill the top-right quadrant.

Step 5: Add Sheet 3 —
a) From the left Sheets panel, drag Sheet 3 into the canvas.
b) Drop it in the bottom-left area of the dashboard.
c) Resize to fill the bottom-left quadrant.

Step 6: Publish the Dashboard to Tableau Public
a) Click File from the top menu → select Save to Tableau Public.
b) Sign in with your Tableau Public credentials if prompted.
c) Name the workbook — ev analysis keep existing name.
d) Click Save — Tableau will upload and open the live link in your browser.
e) Your Dashboard now live at: [ev analysis | Tableau Public]





5) **Storyboard**

A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

Step 1: Create New Story

a) At the bottom of Tableau, click the New Story button next to the sheet tabs.

b) A blank story canvas will open named Story 1.

Step 2: Set Size

a) On the left panel under Size → select Automatic or Desktop Browser (1016 x 964).

Step 3: Add 10 Story Points

a) Click Blank on the left panel to create each new story point.

b) Drag and drop each sheet from the left Sheets panel onto the canvas one by one.

c) After dropping each sheet, click Add a Caption at the top and type the caption name.

Step 4: Add Story Title

a) Check Show Title at the bottom-left panel.

b) Double-click the title and type — Story of Electric Cars In India

Step 5: Publish

a) Click File → Save to Tableau Public.
b) Sign in → Name the workbook → Click Save.
c) Story goes live on your Tableau Public profile.

Story | Tableau Public

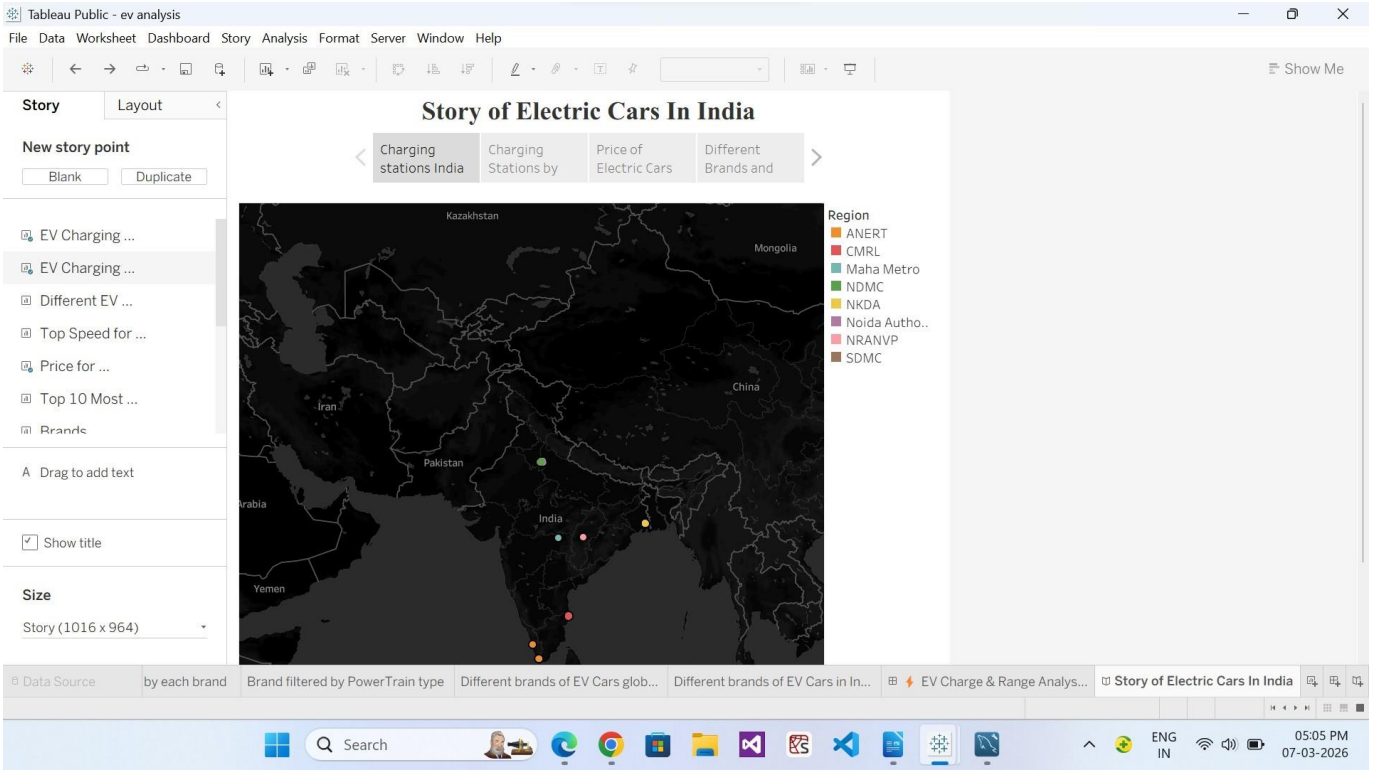


Tableau Public - ev analysis
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
Layout
New story point
Blank
Duplicate
EV Charging ...
EV Charging ...
Different EV ...
Top Speed for ...
Price for ...
Top 10 Most ...
Brands
Drag to add text
Show title
Size
Story (1016 x 964)
Story of Electric Cars In India
Charging stations India
Charging Stations by
Price of Electric Cars
Different Brands and
Kazakhstan
Mongolia
China
Iran
Pakistan
India
Yemen
Region
ANERT
CMRL
Maha Metro
NDMC
NKDA
Noida Autho..
NRANVP
SDMC
Data Source
by each brand
Brand filtered by PowerTrain type
Different brands of EV Cars glob...
Different brands of EV Cars in In...
EV Charge & Range Analys...
Story of Electric Cars In India
Search
ENG
IN
05:05 PM
07-03-2026

6) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

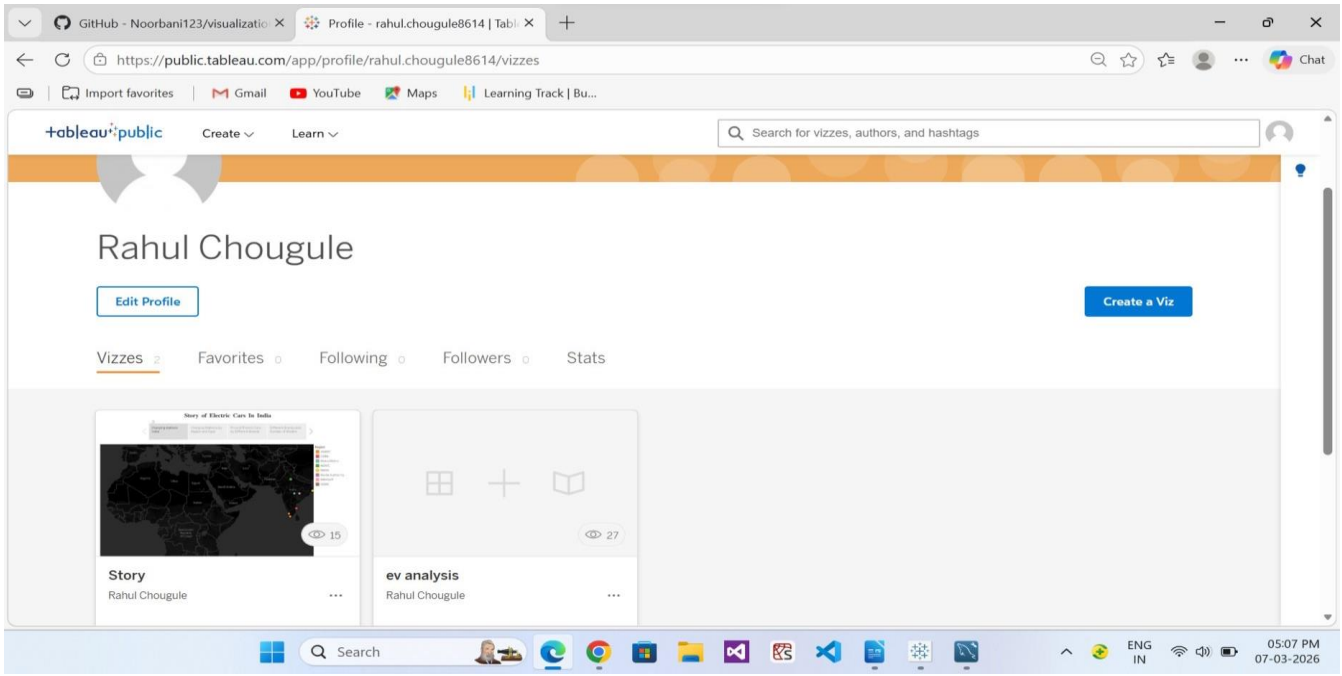
❖ Publishing

a) **Sign in to Tableau Public**

In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.

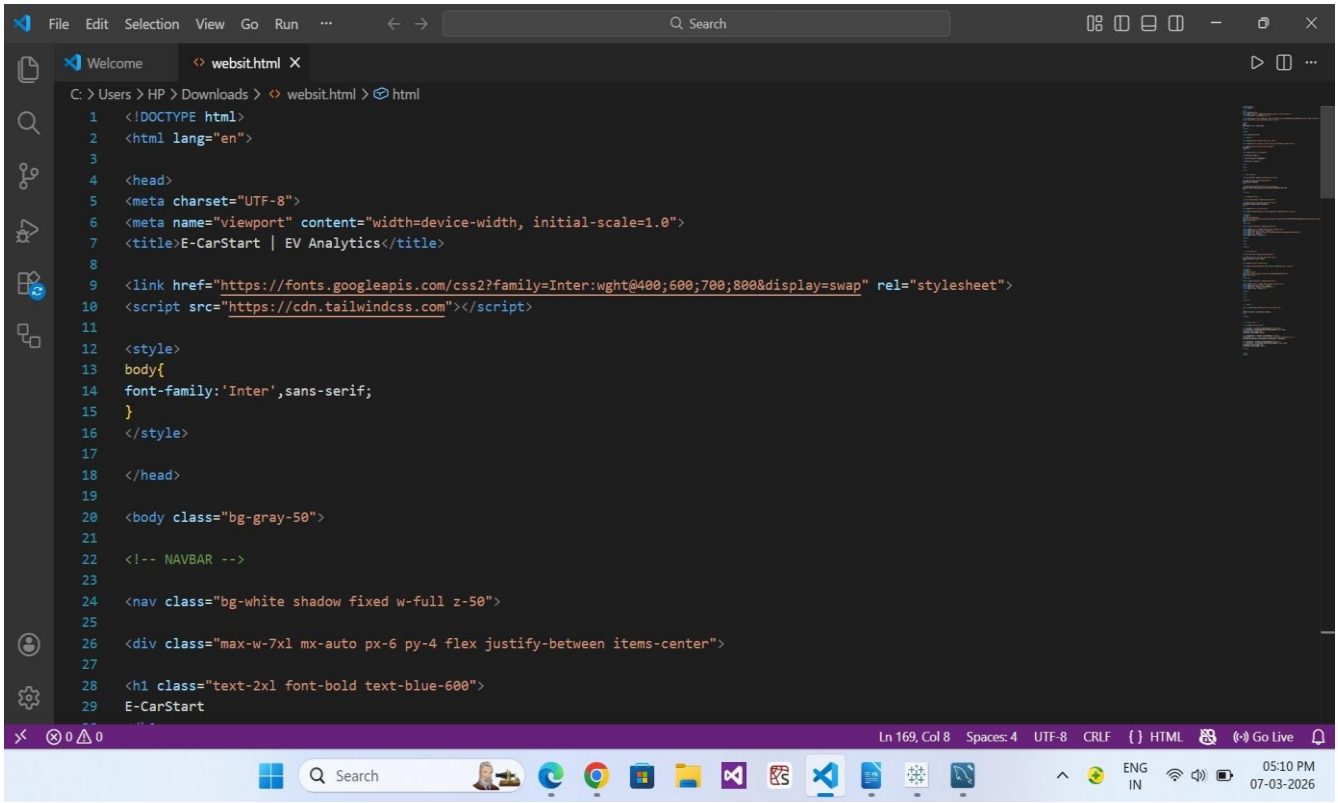


Dashboard and Story embed with UI with Flask.

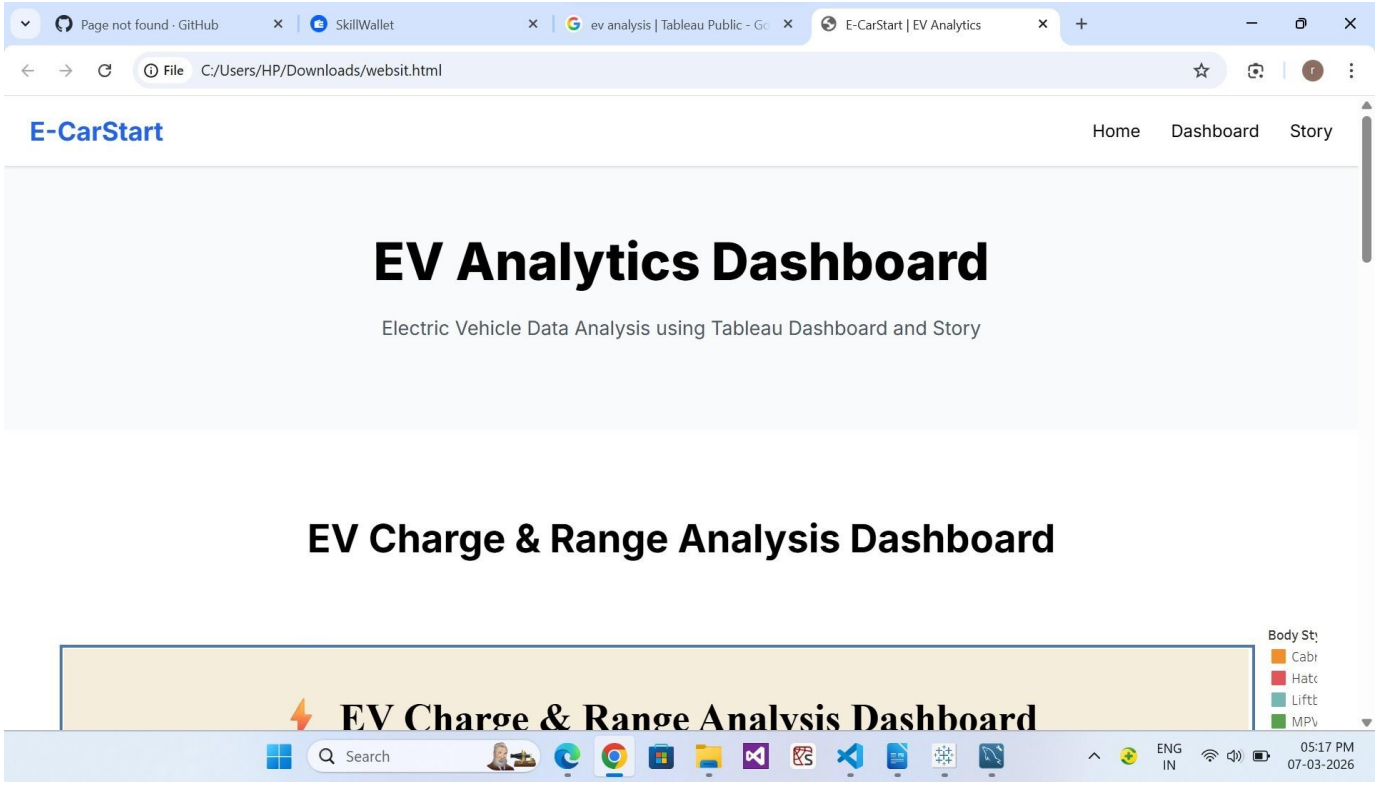
In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

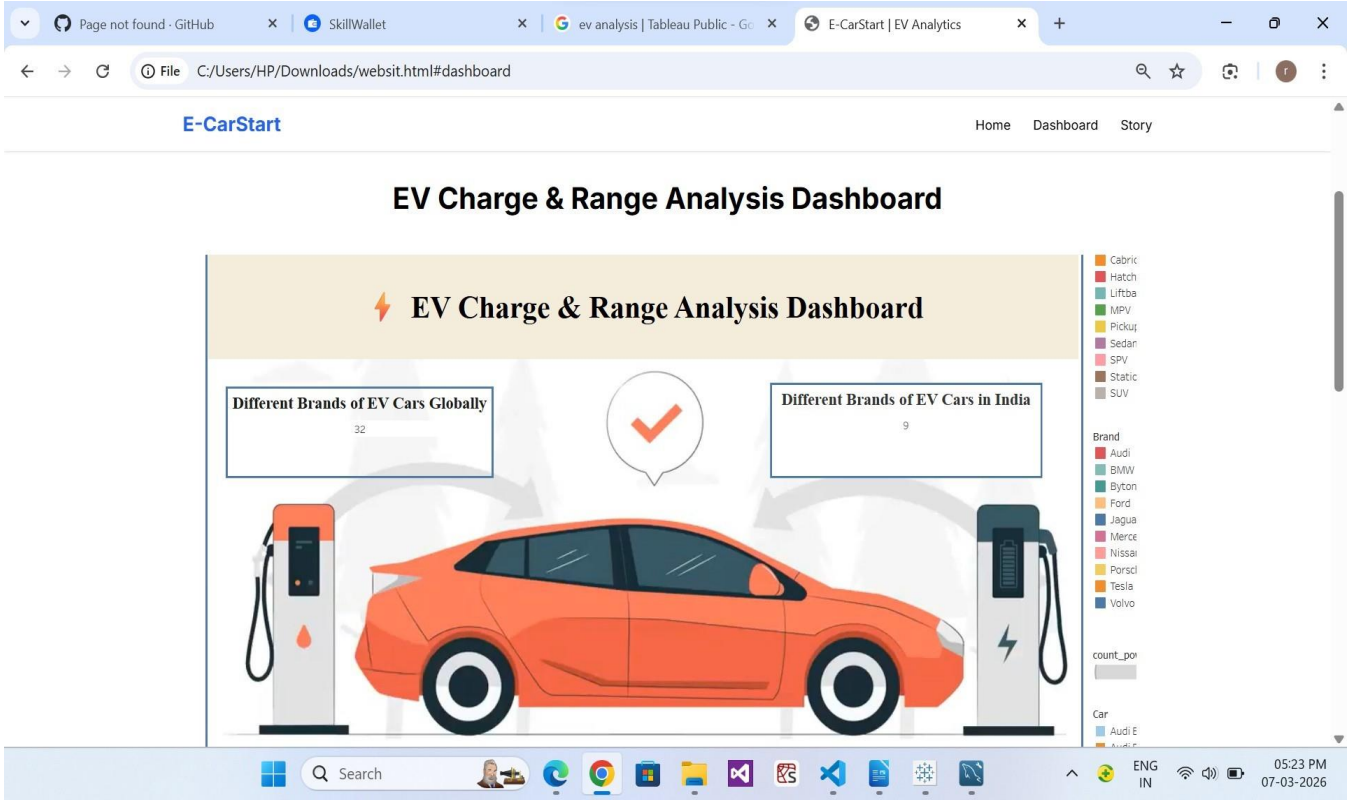
1) Adding Dashboard and Story on HTML page.

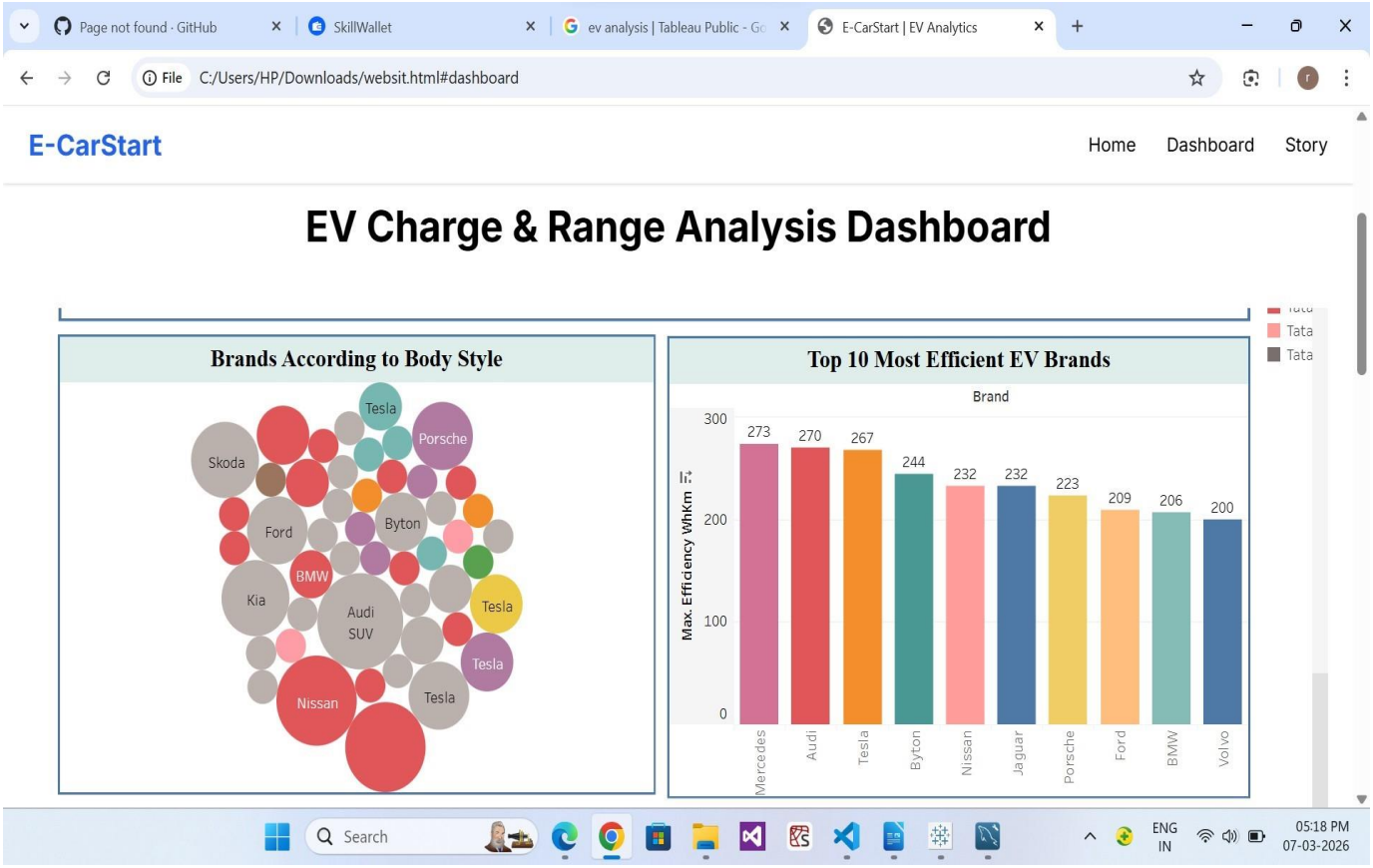
a) Open Tableau Public profile and open Dashboard/Story.
b) Click on share option and copy the embed code.
c) Paste in HTML page

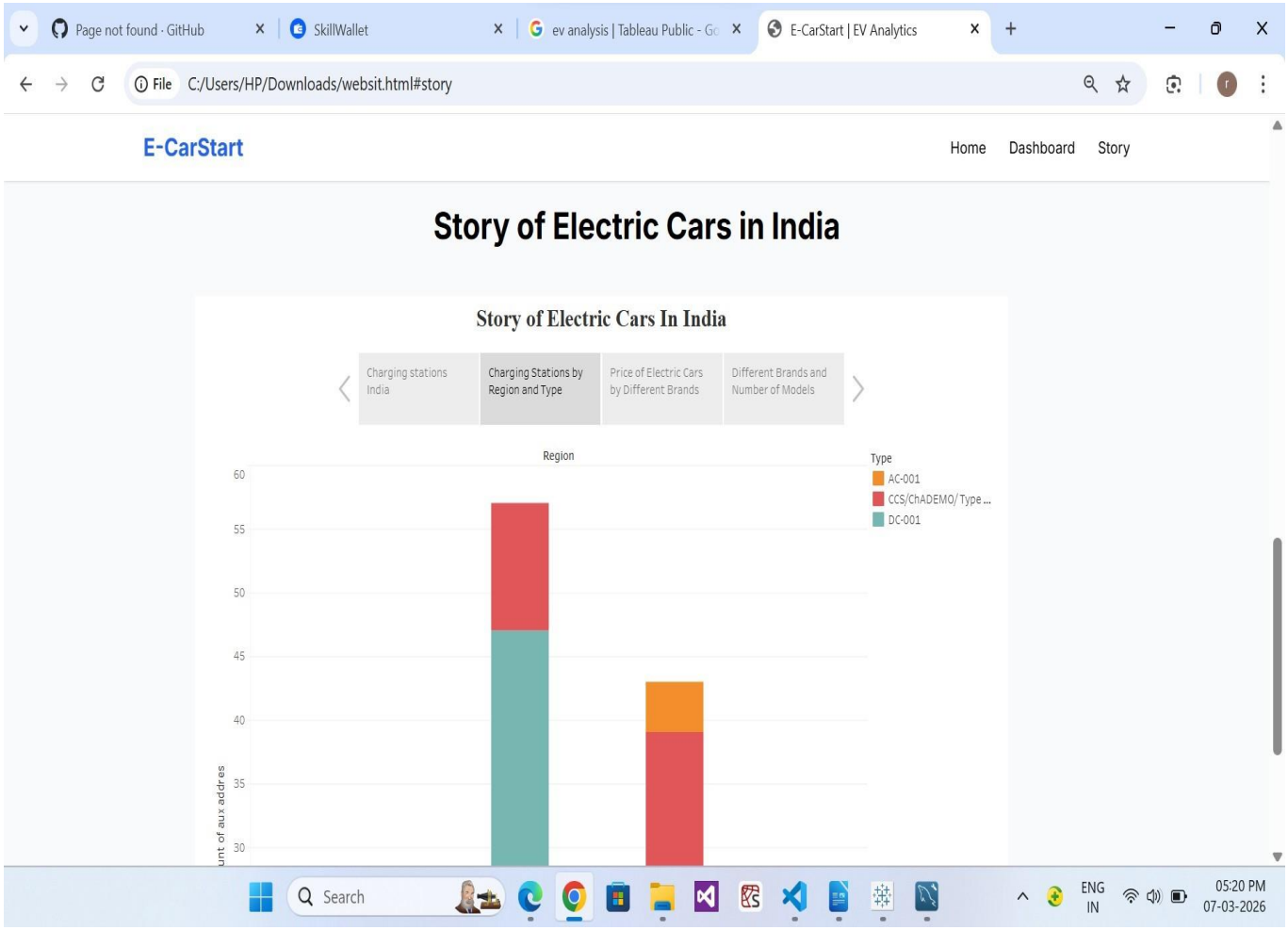


D) Save the .html file and Run the file in Browser.


Page not found · GitHub
SkillWallet
ev analysis | Tableau Public - Go
E-CarStart | EV Analytics
File C:/Users/HP/Downloads/websit.html
E-CarStart
Home
Dashboard
Story
EV Analytics Dashboard
Electric Vehicle Data Analysis using Tableau Dashboard and Story
EV Charge & Range Analysis Dashboard
EV Charge & Range Analysis Dashboard
Search
ENG IN
05:17 PM
07-03-2026


Page not found · GitHub
SkillWallet
ev analysis | Tableau Public - Go
E-CarStart | EV Analytics
File C:/Users/HP/Downloads/websit.html#dashboard
E-CarStart
Home
Dashboard
Story
EV Charge & Range Analysis Dashboard
EV Charge & Range Analysis Dashboard
Different Brands of EV Cars Globally
32
Different Brands of EV Cars in India
9
Brand
Audi
BMW
Ford
Tesla
Volvo
Car
Search
ENG IN
05:23 PM
07-03-2026


Page not found · GitHub
SkillWallet
ev analysis | Tableau Public - Go
E-CarStart | EV Analytics
File C:/Users/HP/Downloads/websit.html#dashboard
E-CarStart
Home
Dashboard
Story
EV Charge & Range Analysis Dashboard
Brands According to Body Style
Tesla
Porsche
Skoda
Ford
Byton
BMW
Kia
Audi SUV
Tesla
Tesla
Nissan
Tesla
Top 10 Most Efficient EV Brands
Brand
Max. Efficiency WhKm
300
200
100
0
273
270
267
244
232
232
223
209
206
200
Mercedes
Audi
Tesla
Byton
Nissan
Jaguar
Porsche
Ford
BMW
Volvo
Tata
Tata
Search
ENG IN
05:18 PM
07-03-2026


Page not found · GitHub
SkillWallet
ev analysis | Tableau Public - Go
E-CarStart | EV Analytics
File C:/Users/HP/Downloads/websit.html#story
E-CarStart
Home
Dashboard
Story
Story of Electric Cars in India
Story of Electric Cars In India
Charging stations India
Charging Stations by Region and Type
Price of Electric Cars by Different Brands
Different Brands and Number of Models
Region
Type
AC-001
CCS/ChADEMO/Type ...
DC-001
60
55
50
45
40
35
30
Search
ENG IN
05:20 PM
07-03-2026